Unit 4
Network Analysis
Lesson 2

Shortest Route/Path Problem

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Shortest Route/Path Problem

Shortest route/path problem is the problem of finding a path between two vertices (or nodes) such that the sum of the weights of its constituent edges is minimized. It is used to determine route between a source and destination in a transportation problem.

An example is finding the quickest way to get from one location to another on a road map; in this case, the vertices represent locations, and the edges represent segments of road and are weighted by the time needed to travel that segment.

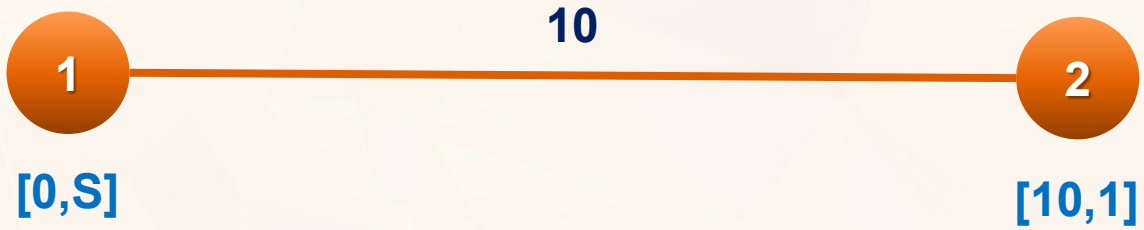
Shortest Route/Path Problem Algorithm

1. Label the source node **(starting node)** with the permanent [0,S]. The leading number is the distance from node 1; the second number is the following node. Because node 1 has no preceding node, it is assigned a label S for the starting node.

2. Compute tentative labels, **(d,n),** for the nodes that can be reached directly from node 1. d = the direct distance from node 1 to the node in question -- this is called the distance value. n indicates the preceding node on the route from node 1 -- this is called the preceding node value.

3. If all nodes have permanent labels, **stop.** Otherwise, select the label with the shortest distance from all temporary labels.

Example 1:

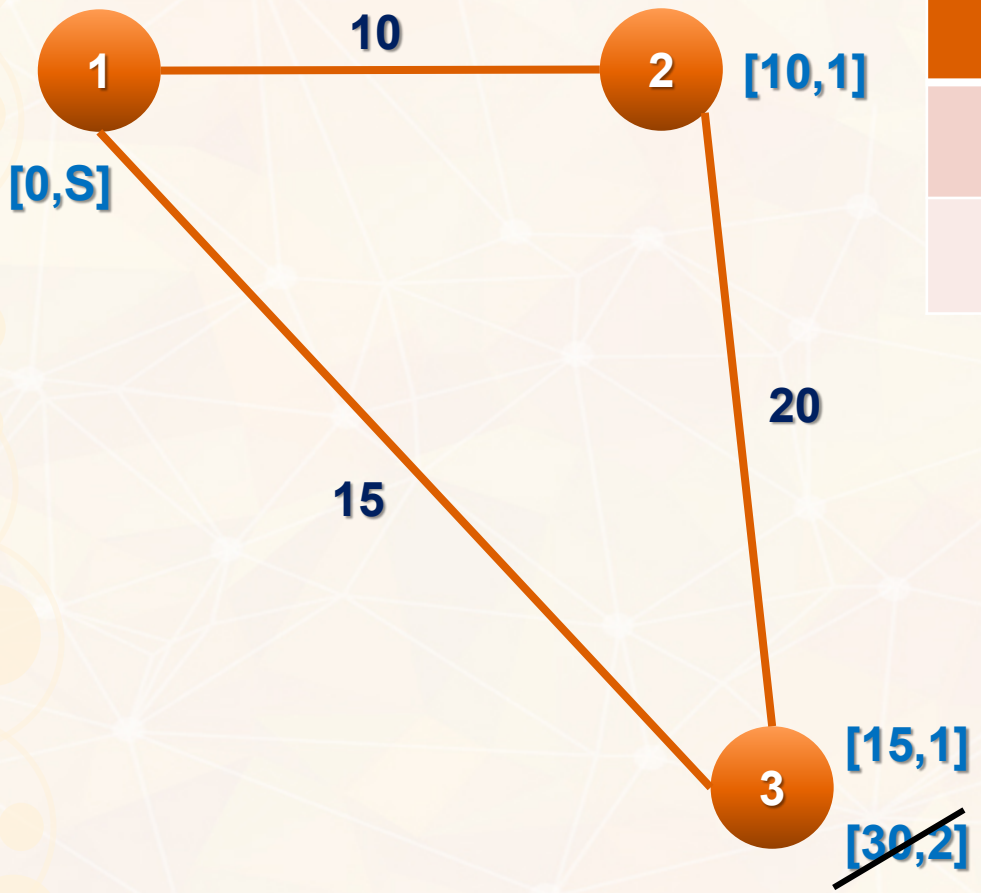


Example 2:



Example 3:

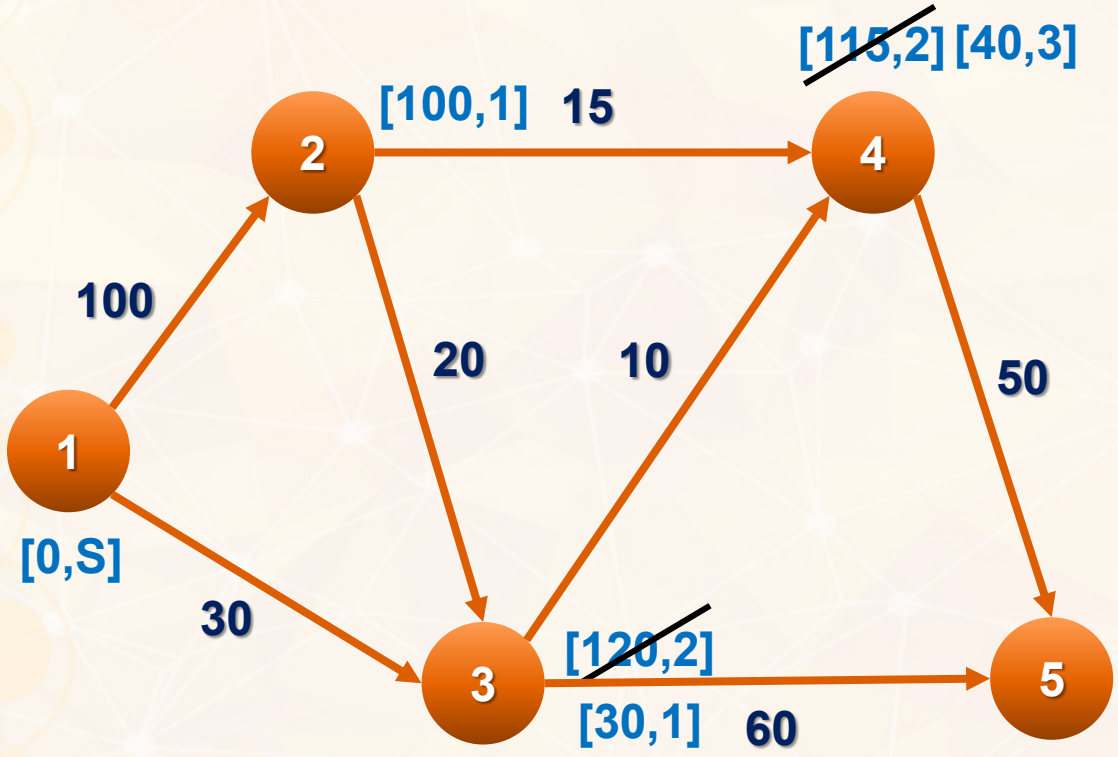
Find the shortest route from Node 1 to all other nodes in the network:



Destination (Node)	Minimum Distance	Shortest Route/Path
2	10	1-2
3	15	1-3

Example 4:

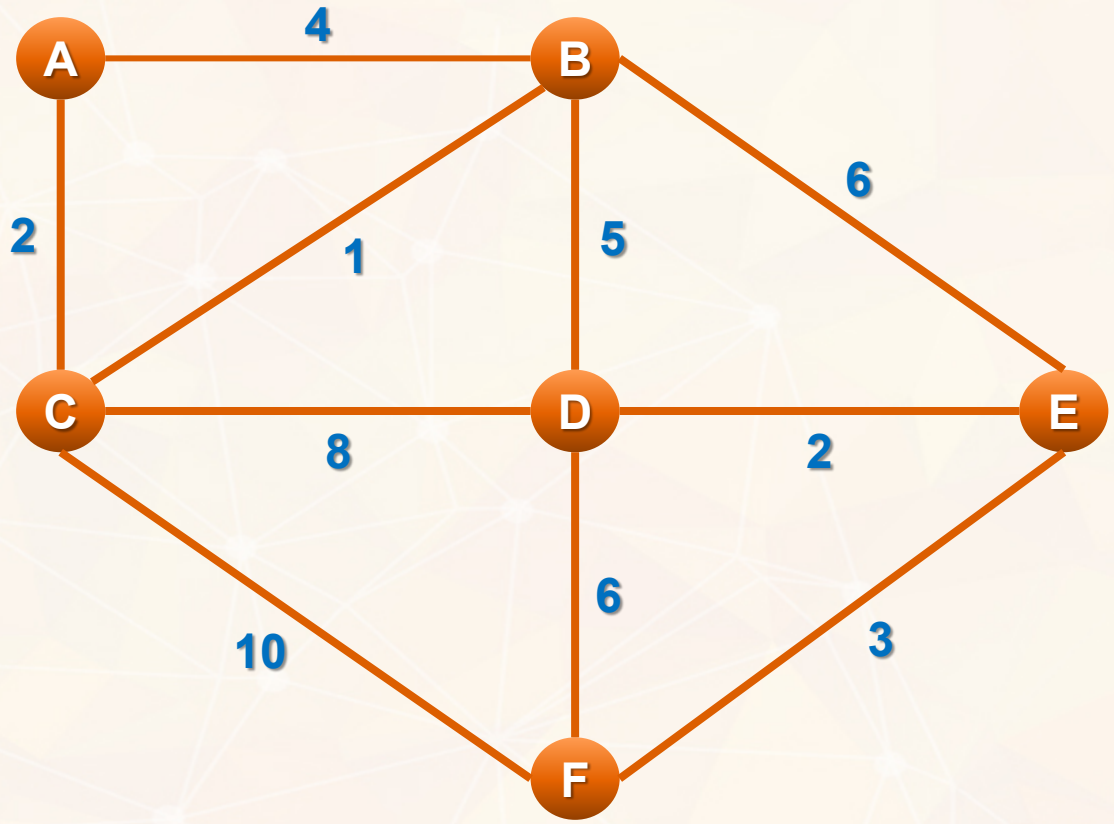
Find the shortest route from Node 1 to all other nodes in the network:



Destination (Node)	Minimum Distance	Shortest Route/Path
2	100	1 - 2
3	30	1 - 3
4	40	1 - 3 - 4
5	90	1 - 3 - 4 - 5 1 - 3 - 5

Lesson Activity:

Case Scenario: Starlight Delivery Company needs to transport a package from Warehouse A to Warehouse F. The network below represents the possible roads between locations. The numbers on the branches represent the distance in kilometers (km). Determine the shortest route between the nodes. If Starlight Delivery Company charges ₱50 per kilometer, how much will it cost to travel using the shortest route?



Assessment Criteria

Total Points: 100 points

Scale: 1–5 (20 points per criterion)

Criterion 1: Identification of Routes

Criterion 2: Calculation of Total Distance

Criterion 3: Determination of Shortest Route

Criterion 4: Solution Process and Group Participation

Criterion 5: Accuracy, Organization and Presentation

Note:

Team Leader nalang ang magsusubmit/turn in ng activity

Keep safe always, stay healthy and God bless you all! ♥♥♥

Thanks for listening!

God Bless You!!!